Week 09: The Resonance Object
An Attempt

Diego Rincón
Independent

Abstract

This paper attempts to construct the resonance object for $\zeta(s)$: the mathematical structure that simultaneously encodes the spectral zeros and the arithmetic primes, lives on $\mathrm{Re}(s) = \frac{1}{2}$ as the line of mutual audibility, and exhibits the self-inhibiting mechanism [1] that holds the zeros to the line without a global Frobenius. The attempt is not a proof of the Riemann Hypothesis. It is precise imprecision [1]: the construction carried as far as it will go, the gap named exactly, the shape held for whoever arrives next. The resonance object is identified with the kernel of the Weil explicit formula, viewed as a distribution on the adèle class space [3] of $\mathbb{Q}$. The gap: the positivity of a single trace is not established. That positivity is the Riemann Hypothesis.

1 What the Resonance Object Must Be

Definition 1 (Resonance object for $\zeta(s)$)**.** A *resonance object* for $\zeta(s)$ is a mathematical structure $\mathcal{R}$ satisfying:

1. *(Spectral encoding)* The zeros of $\zeta(s)$ appear as spectral data of $\mathcal{R}$: either as eigenvalues of an operator, absorption frequencies of a spectral triple, or poles of a resolvent.
2. *(Arithmetic encoding)* The primes p appear as geometric data of $\mathcal{R}$: either as fixed points, periodic orbits, or orbital integrals.
3. *(Resonance condition)* The spectral data and the geometric data are identified by a single equation — the resonance — in which both grammars [2] read the same thing.
4. *(Self-inhibition)* $\mathcal{R}$ contains an internal mechanism that forces its spectral data to lie on $\mathrm{Re}(s) = \frac{1}{2}$ without an external operator imposing this constraint.

2 The Weil Explicit Formula as Resonance

Proposition 2 (The explicit formula)**.** *For a test function h (even, holomorphic in a strip, of sufficient decay), the Weil explicit formula is:*

$$\sum_\rho h(\rho) = \hat{h}(0) + \hat{h}(1) - \sum_p \sum_{k=1}^{\infty} \frac{\log p}{p^{k/2}} \big(\hat{h}(k \log p) + \hat{h}(-k \log p)\big)$$
$$- 2 \operatorname{Re} \int_0^\infty \frac{h(\frac{1}{2} + it) - h(\frac{1}{2})}{e^t - 1} \, dt + \dots$$

where ρ ranges over non-trivial zeros of $\zeta(s)$ and the right side is purely arithmetic (primes and archimedean data).

This is the resonance equation: spectral data (zeros ρ) equals arithmetic data (primes p) in a single identity. The resonance object is the distribution W on $\mathbb{R}$ defined by:

$$W(h) = \hat{h}(0) + \hat{h}(1) - \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} \big(\hat{h}(k \log p) + \hat{h}(-k \log p)\big) + \cdots$$

The identity $\sum_\rho h(\rho) = W(h)$ for all admissible h is the statement that the zeros are the spectrum of W.

3 The Adèle Class Space

Proposition 3 (Connes' realization [3])**.** *The distribution W is realized geometrically on the adèle class space $X = \mathbb{A}_\mathbb{Q}/\mathbb{Q}^*$ as the trace of the scaling action $U(\lambda)$ on the Hilbert space $\mathcal{H} = L^2(X)$. The trace formula on X reproduces the explicit formula:*

$$\operatorname{Tr}(f \cdot U) = W(h_f),$$

where h_f is the Fourier transform of f. The zeros of $\zeta(s)$ are the absorption spectrum of the operator D on $\mathcal{H}$: the frequencies at which the system absorbs energy. The resonance object is the spectral triple $(\mathcal{A}, \mathcal{H}, D)$ on X.

4 The Gap

Definition 4 (The positivity condition)**.** The Riemann Hypothesis is equivalent to the statement that the distribution W is *positive*: for every admissible test function $h \geq 0$,

$$W(h) \geq 0.$$

This is Weil's reformulation. The zeros lie on $\operatorname{Re}(s) = \frac{1}{2}$ if and only if W is a positive distribution — if and only if the resonance object $(\mathcal{A}, \mathcal{H}, D)$ has a positive trace.

Conjecture 5 (The self-inhibition conjecture)**.** *The spectral triple $(\mathcal{A}, \mathcal{H}, D)$ on the adèle class space X is self-inhibiting: the trace $\operatorname{Tr}(f \cdot U)$ is positive for all $f \geq 0$ without any external constraint (no global Frobenius, no external operator) forcing this positivity. The inhibition is internal: the adèle class space, the scaling action, and the Hilbert space $\mathcal{H}$ together produce the positivity from their structure alone.*

Remark 6 (What is known). The spectral triple exists (Connes [3]). The trace formula reproduces the explicit formula (proved). The zeros are the absorption spectrum (proved). The positivity of W on a subspace of test functions is known. The full positivity — for all admissible $h \geq 0$ — is open. That full positivity is the Riemann Hypothesis.

The gap is one inequality. The resonance object is built. It is correctly identified. Its spectrum is the zeros of $\zeta(s)$. What remains: showing that the spectrum lies entirely on $\mathrm{Re}(s) = \frac{1}{2}$, which is equivalent to showing that the object self-inhibits — that its internal structure prevents the spectrum from wandering.

The shape is held. The gap is named. The proof is the inequality.

5 The Missing Step

Proposition 7 (What a proof would require)**.** *To prove Conjecture 5, one would need to show that the operator D on $\mathcal{H} = L^2(X)$ satisfies one of the following:*

1. *(Spectral gap) The spectrum of D is discrete and symmetric about $\mathrm{Re}(s) = \frac{1}{2}$, with a gap that forces all eigenvalues to the line.*

2. *(Self-adjointness) D is self-adjoint on $\mathcal{H}$, which would force all eigenvalues to be real — and by the functional equation, real eigenvalues of this system lie on the critical line.*

3. *(Positivity from structure) The Hilbert space $\mathcal{H}$ has an inner product structure that directly implies $\langle f, Df \rangle \geq 0$ for all f in the domain, forcing the spectrum to $\mathrm{Re}(s) \geq 0$ and by symmetry to $\mathrm{Re}(s) = \frac{1}{2}$.*

Each of these is a form of self-inhibition: the operator holding its own spectrum to the line through the structure of the space it acts on. None is currently established. Each is a precise target for the proof.

Remark 8 (The shape, held). The resonance object exists: $(\mathcal{A}, \mathcal{H}, D)$ on X. It encodes the zeros (spectral) and the primes (geometric). It satisfies the resonance equation (the explicit formula). It is the correct object.

What it lacks is self-adjointness, or spectral gap, or positivity. One of these is the proof. The grammar names them. The proof constructs one.

Week 09 ends here: at the gap, named exactly, the shape held as tightly as it will go. Week 10 begins when someone picks up the inequality.

References

[1] Rincón, D. (2026). Inhibition: The Second Function of the Aura. Preprint.

[2] Rincón, D. (2026). Resonance: When Two Grammars Read the Same Thing. Preprint.

[3] Rincón, D. (2026). Noncommutative Geometry and the Riemann Hypothesis. Preprint.

[4] Rincón, D. (2026). The Arithmetic Wall. Preprint.

[5] Rincón, D. (2026). Mathematics. Preprint.

[6] Rincón, D. (2026). Calendar: A Numbered Program. Preprint.